DAILYN DESPRADEL RUMALDO

XR Research Scientist — UX Researcher — Product Design — Accessibility

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Summary

Doctoral researcher, XR interaction developer, and human-centered technologist specializing in ***accessible and adaptive virtual, augmented,*** and ***mixed-reality systems***. Experienced in developing real-time multimodal interactions using *EMG, eye gaze, head movement, hand tracking, controllers, motion capture, and physiological sensing*. Combines user research, *machine learning, biomechanics, signal processing*, and *3D prototyping to create and evaluate interactive systems*. Industry experience includes ergonomics and product-design research at **Apple** and intent-aware robotic-system development at **Meta**.

Education

Carnegie Mellon University (CMU), Pittsburgh, PA **GPA: 3.77/4.00**

Ph.D. in Mechanical Engineering,Human-Computer Interaction Focus Anticipated Graduation: Dec 2026

Dissertation: Toward Adaptive and Accessible Extended Reality Interactions: Understanding and Enhancing User Experience

New York Institute of Technology (NYIT), New York City, NY **GPA: 3.66/4.00**

M.S. in Electrical and Computer Engineering Graduation: May 2021

Mercy College, Dobbs Ferry, NY **GPA: 3.71/4.00**

B.S. in Biology — *McNair Scholar* Graduation: May 2018

Industry Experience

Research Scientist Intern, Meta Inc. **Oct 2025 - March 2026**

User Experience and Robotics Systems

- Led the development of an **end-to-end sEMG-enabled robotic teleoperation pipeline**, collaborating with five cross-functional Meta teams to integrate wearable sensing, XR interaction, and robot control.
- **Built immersive XR environments** in Unreal Engine using Python, Unreal Engine Blueprints, and SolidWorks to investigate sEMG-derived force as a continuous input for dexterous robotic hand teleoperation.
- **Planned and executed an internal user study** with 12 Meta employees, overseeing participant recruitment, experimental design, data collection, quantitative analysis, and **presentation of findings to multidisciplinary stakeholders**.

Vision Pro Group (VPG) Product Design Intern, Apple Inc. **April 2025 - September 2025**

Ergonomics / Product Design Analyst Intern

- **Conducted ergonomic evaluations and competitive benchmarking of XR hardware**, leveraging user studies, optical capture, and 3D scanning to assess comfort, fit, placement, wearability, and usability across diverse users.
- **Analyzed large-scale 3D scan data**, ergonomic measurements, survey responses, and existing user-research datasets using Python and Siemens NX to identify trends and deliver evidence-based product design recommendations.
- Led an independent **ergonomics research project by defining objectives, assembling and coordinating a cross-functional team, and presenting technical findings that informed product design, engineering, and human factors decisions**.

PhD Research Experience

Research Lead — Human–Computer Interaction, Accessibility & Extended Reality **August 2021 – Present**
NeuroMechatronics Lab, CMU (PI: Dr. Douglas Weber)

Research Focus: Adaptive XR Interactions: Understanding and Enhancing User Experience

Neuromechanics of Object Manipulation Across Physical and Virtual Environments 2021 – 2022

- Designed and conducted human-subject studies comparing object manipulation across physical, haptic-free virtual, and haptic-enabled virtual environments using Unity, motion capture, hand kinematics, electromyography (EMG), and wearable haptic devices.
- Developed experimental and data analysis pipelines to quantify neuromuscular activity and movement behavior during pre- and post-contact phases of object manipulation, identifying differences in interaction strategies across physical and virtual environments.
- Established a quantitative framework for evaluating VR interaction fidelity and the effects of haptic feedback on object manipulation, publicly documenting the research methodology, findings, and implementation through an open-source GitHub repository.

Accessible Human–Computer Interaction Using Wearable sEMG (Meta Collaboration) 2022 – Present

- Led a collaborative CMU–Meta research program developing Unity-based interaction environments that enabled people with tetraplegia to control computers and XR applications using wearable surface electromyography (sEMG), resulting in a first-author manuscript under review at *Science Robotics*.
- Designed and conducted longitudinal human-subject studies with individuals with cervical spinal cord injury, developing real-time motor-unit decoding pipelines for cursor control, object selection, continuous interaction, and immersive XR experiences.
- Extended the work to virtual reality by developing Meta Quest Pro assessment environments that quantified cervical mobility, head movement, and comfortable visual workspaces, establishing participant-specific accessibility metrics that informed personalized XR interaction design.

Adaptive, Force-Aware XR Interfaces for Accessibility *2025 – Present*

- Developed participant-specific machine learning pipelines using multilayer perceptrons (MLPs) to estimate continuous hand force from wearable sEMG, enabling real-time force-responsive interaction in immersive XR.
- Designed multimodal XR interaction techniques integrating eye gaze, head movement, physiological sensing, cervical mobility assessment, and constrained optimization to personalize interface placement based on user capability and environmental context.
- Developing a shared-autonomy framework that recommends personalized XR interface layouts using user mobility, force estimation, environmental context, and continuous user feedback to improve accessibility for people with spinal cord injury, wheelchair users, and individuals experiencing situational mobility limitations.

Selected XR Projects

Neuromechanics of Interaction | *Unity, VR, EMG, Motion Capture, Haptics* **Repository**

- Developed physical, haptic-free VR, and haptic-influenced object-manipulation experiments to quantify changes in hand kinematics, muscle activity, and interaction behavior.

Clean the Dishes VR | *Unity, Eye/Head Tracking, Controllers, EMG* **Repository**

- Built a multimodal VR task supporting interchangeable gaze, head-directed, controller, and EMG-based input for object selection and placement.

Peg-and-Hole Force Task | *Unity, Force Input, Hand Tracking, Data Logging* **Repository**

- Developed a force-sensitive motor-control experiment with randomized task configurations, force constraints, hand-joint tracking, and real-time experimental logging.

How Do You See the World? | *Unity, VR Accessibility, Post-Processing* **Repository**

- Created an immersive experience simulating cataracts, glaucoma, and color-vision deficiencies across everyday virtual environments.

Ultraleap Mid-Air Interaction | *Unity, Ultraleap, Hand Tracking* **Repository**

- Built controller-free desktop interaction environments using mid-air hand tracking and gesture-based object manipulation without an HMD.

Inclusive Hand Animation Library | *Blender, Modeling, Rigging, Animation*

- Created diverse rigged and animated hand assets for inclusive XR prototyping, accessibility research, and interaction-design demonstrations.

Master's Research Experience

Research Lead - VR and EMG-Based Interaction for Prosthetic Rehabilitation **Aug 2019 – May 2021**
NYIT (Advisor: Dr. N. Sertac Artan)

- Developed VR rehabilitation interactions using Oculus Quest 2 and the Myo Armband (EMG) for gesture-based virtual arm control, evaluating real-time control of four gestures (fist, open hand, wrist flexion/extension).
- Analyzed latency and responsiveness between EMG sensing and VR interaction pipelines to improve usability of assistive XR applications for prosthetic rehabilitation

Technical Expertise

XR & Interactive Development: Unity, Unreal Engine, AR, VR, Mixed Reality, WebXR, C#, Unreal Blueprints
User Experience (UX) Research: Eye Tracking, Hand Tracking, EMG, Motor Unit Interfaces, Motion Capture, Accessibility, Human Factors, Ergonomics, User Experience Research
Programming & Data: Python, Signal Processing, Machine Learning, Data Visualization, Experimental Design, Statistical Analysis, Sensor Fusion
3D Design & Prototyping: Blender, Siemens NX, Fusion 360, SolidWorks, 3D Scanning, Optical Measurement, Animation, Rigging
Languages: Spanish (Native), English (Professional Fluency)

Fellowships

- GEM Fellow, CMU, Mechanical Engineering, 2021 - Present
- McNair Scholar, Mercy College, Biology, 2018

Selected Publications and Academic Output

- **D. Despradel**, et al. “A Wearable Myoelectric Sensor Enables Human–Machine Interaction for People with Tetraplegia.” *Science Robotics*, Under Review.
- **D. Despradel**, et al. “Enabling Advanced Interactions through Closed-Loop Control of Motor Unit Activity After Tetraplegia.” *ACM Symposium on User Interface Software and Technology (UIST)*, Pittsburgh, PA, 2024.
- **D. Despradel**, et al. “Closed-Loop Control of Motor Unit Activity After Tetraplegia for Human-Computer Interaction.” *Society for Neuroscience Annual Meeting*, Chicago, IL, 2024.
- **D. Despradel**, et al. “Real-Time Detection and Decoding of Motor Unit Activity Using a Wearable Neuromotor Interface in a Person with Motor Complete Tetraplegia.” *Neural Control of Movement*, Dubrovnik, Croatia, 2024.

Additional presentations available upon request.

Leadership, Outreach & Lab Development

- **Social Media Manager, NeuroMechatronics Laboratory** — Created research highlights, outreach content, and student spotlights to increase the visibility of the lab’s research.
- **Motion-Tracking Laboratory Development** — Led the design and development of a dedicated motion-tracking research space supporting biomechanics, XR, and human-subject experiments.
- **Black History Month VR Event (CMU)** — Led an immersive VR experience (*MLK: Now Is the Time*) for the College of Engineering DEI program, 2024.
- **Biomechanics Day Outreach** — Led K–12 STEM workshops on biomechanics, Blender animation, and human movement, 2023–2024.
- **Mentorship Chair, MEGSO** — Organized graduate student mentorship initiatives and community engagement, 2023–2024.